Research Task: Hybrid Retrieval in the Real World

Oct 13, 2026

Purpose and timeline

In 60-ish minutes, find out how real search systems combine keyword and vector search, compare their choices with ours, and bring back one recommendation backed by a source.

You produce a one-page briefing, using the template near the end, and a 2-minute share-back. Work in teams of two or three. Each team takes one search system.

Time	What you do
0:00 to 0:05	Read this brief, form teams, pick your system
0:05 to 0:25	Core questions: everyone answers these
0:25 to 0:40	Your system: how it does hybrid search
0:40 to 1:00	Write the one-pager, including the comparison with our app
Wednesday	Share back: one finding, one surprise, one recommendation

Where our app is now

Our app searches by meaning only. It has no keyword search, so hybrid retrieval is what we are researching today, not something we run.

Part	What we have today	Where
Retrieval	Vector (semantic) search only: Voyage embeddings stored in Chroma, ranked by cosine similarity	`chunk_index.py`, `knowledge_store.py`

Part	What we have today	Where
Chunking	`sections_contextual`: one chunk per section, each labelled "Title > Heading"	`chunking.py`
Chunks sent to the model	The top 3	`/knowledge/ask`
Relevance gate	Refuse unless a chunk scores a cosine of at least 0.45	`routers/knowledge.py`
Keyword search	None	

The gap that brings us here: `q24`, "What is matter LP-3307?", is refused at every relevance floor we tried. The likely reason is that a code like LP-3307 means nothing to an embedding model, so its chunk scores low. Keyword search would match `3307` exactly. Your job is to find out how real systems combine the two, and whether we should.

The research questions

Every team answers the three core questions, then researches one system in depth. Write the source link next to every answer.

Core questions (everyone, 15 minutes)

1. **Rank or score?** There are two families of fusion. Rank-based fusion (such as Reciprocal Rank Fusion) uses only each list's order. Score-based fusion rescales each list's scores and adds them. What does each gain, and what does each throw away?
2. **Does hybrid actually help?** Find one published comparison with numbers: hybrid against keyword-only or vector-only. Note the dataset, the metric, the size of the gain, and who ran it.
3. **Where does the keyword half come from?** In production, is it classic BM25 from an inverted index, or a learned sparse model? Find one example of each.

Your system (one per team, 25 minutes)

Team	System	A question to start from
A	Elasticsearch	Its RRF has a weight per retriever. When would you set one above 1.0?
B	Weaviate	Its default fusion is score-based, with an `alpha` setting. Why might it prefer that to rank-based fusion?
C	Qdrant	It offers two fusion methods. What does the second one do with each list's scores?
D	Azure AI Search	It always fuses hybrid results with RRF, then can rerank them. What runs after fusion, and why?
E	Postgres with pgvector	There is no hybrid switch. How would you write it yourself in SQL?
F	Chroma	It has a hybrid Search API. Can our local, persistent Chroma use it, or only Chroma Cloud?

For your system, find out:

- How you ask for a hybrid search: the query, setting or API.
- Which fusion method it uses, and its default settings.
- Whether you can weight one retriever above the other, and how.
- How it builds the keyword side: BM25, sparse vectors, or something else.
- One limit or catch: a version requirement, a cloud-only feature, a cost.
- What its documentation says about testing whether hybrid helps.

Starter sources

Start from these, but go beyond them...

Source	From	Use it for
Reciprocal Rank Fusion outperforms Condorcet and individual rank learning methods	Cormack, Clarke and Buettcher, SIGIR 2009	Where RRF and k = 60 come from. Everyone
RRF retriever	Elastic documentation	Team A: parameters, defaults, weights, a hybrid example
Practical BM25, part 3: picking b and k1	Elastic blog	Everyone: what the BM25 settings do
Hybrid search concepts	Weaviate documentation	Team B: fusion methods and `alpha`
Hybrid search fusion algorithms	Weaviate blog	Team B: rank-based against score-based fusion, with a worked example
Hybrid and multi-stage queries	Qdrant documentation	Team C: `prefetch`, `rrf` and `dbsf`
Hybrid search tuning	Qdrant documentation	Team C and everyone: how to test whether fusion helps
Relevance scoring in hybrid search using RRF	Microsoft Learn	Team D: how Azure AI Search fuses, and what runs after
Outperforming vector search with hybrid retrieval and ranking	Microsoft Tech Community	Core question 2: a published comparison, run by a vendor
Filtering and hybrid search	EDB documentation	Team E: hybrid search written in SQL, with RRF
Build hybrid search with BM25 and vector similarity	Tiger Data documentation	Team E: getting true BM25 into Postgres
Search API overview	Chroma documentation	Team F: what the Search API does, and where it runs

How to research well

Trust the system's own documentation first, check the version, and write a link next to every claim.

- **Rank your sources.** The system's own documentation first, then research papers, then vendor blogs, then everything else.
- **Defaults change between versions.** A real example from preparing this brief: Weaviate's documentation says its default fusion method changed to `relativeScoreFusion` in version 1.24, while a third-party client's guide still labels `rankedFusion` as the default. When sources disagree, trust the vendor and note the version.
- **Ask who ran the benchmark.** A vendor testing its own product is evidence, but it isn't neutral. Note the dataset, the metric, and what was compared.
- **Separate what a source says from what you think.** Write "Qdrant's docs say" or "we think", never both in one sentence.
- **Paraphrase.** Quote a short phrase at most, and link it.
- **Time-box.** If a page hasn't answered your question in three minutes, move on and note the question as open.

What to hand in

One page per team, in this shape. Every claim carries a link.

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# Hybrid retrieval briefing: <system>, Team <letter>

## How it does hybrid search
Two or three sentences. [source](link)

## Fusion and its defaults
Method:          rank-based or score-based, and its name
Defaults:        for example k, and how many results from each list
Weighting:       can one retriever count for more? How?
[source](link)

## The keyword side
BM25, learned sparse vectors, or something else. [source](link)

## One catch
A version, a cloud-only feature, a cost, or a limit. [source](link)
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## Evidence that hybrid helps
Dataset, metric, size of gain, and who ran it. [source](link)

## Core questions
1. Rank or score: what each gains and loses.
2. Does hybrid help: one comparison with numbers.
3. The keyword half: one BM25 example, one learned sparse example.

## Advice for our app
The completed table from "Apply it to our app".
Should we add hybrid search? Yes or no, because ...

## Open questions
What you could not answer in the time.
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